

Do clinical phenotypes reveal heterogeneous benefits of chest radiographs for ICU transfer prediction?

Clinical data–driven phenotyping to evaluate the heterogeneous incremental value of CXR

Model A Clinical data only vs. Model B Clinical data + CXR

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Clinical question

From average model performance to Phenotype-specific value

Why this matters

- Existing early-warning models mainly rely on structured clinical data.
- CXR may provide additional prognostic information for clinical deterioration.

Our central question

Does CXR improve next-day ICU-transfer prediction beyond clinical data, and does its added value differ by clinical phenotype?

Index: non-ICU chest radiograph

Outcome: ICU transfer on the next calendar day

What variables should define the clinical model?

Variable selection grounded in prior ICU deterioration / transfer studies

Evidence	Population / outcome	Variables repeatedly informative	Take-home
Churpek et al. 2014	Ward deterioration / ICU transfer	Vital signs + labs + age	Routine EHR variables can outperform simple early-warning scores
Cheng et al. 2020	Hospitalized COVID-19 / ICU transfer ≤ 24 h	RR, SpO ₂ , WBC, CRP, SBP/DBP, BUN, creatinine, anion gap, lymphocytes	Respiratory, inflammatory, shock and renal signals are important
DeepCARS 2023	General wards / IHCA or unplanned ICU transfer	Age, SBP/DBP, HR, RR, and temperature	AI model outperformed NEWS/MEWS in prospective multicenter validation
eCART comparison 2024	362,926 ward encounters / deterioration ≤ 24 h	Structured EHR early-warning inputs	Modern EHR model achieved AUROC 0.895

→ **Candidate domains for our dataset: demographics + routine CBC/chemistry; add vital signs/oxygenation if reliably available before index CXR.**

Refs: Churpek et al. 2014; Cheng et al. J Clin Med 2020; Cho et al. Crit Care 2023; Edelson et al. JAMA Netw Open 2024.

Why add chest radiographs?

CXR is common, accessible, and may carry complementary physiologic information

Accessible modality

CXR is routinely obtained in ward and emergency care and captures pulmonary and circulatory abnormalities.

Prior evidence

A multimodal clinical+CXR model outperformed a clinical-only model for 30-day ICU mortality
AUC 0.82 vs. 0.72

But “multimodal is better on average” is not enough.

Research gap → Is the incremental value of CXR heterogeneous across clinically distinct patient groups?

Study hypothesis and novelty

The target is heterogeneity of incremental value

H1. The benefit of adding CXR will not be uniform across clinical phenotypes.

MODEL A

Clinical data only

Δ performance

MODEL B

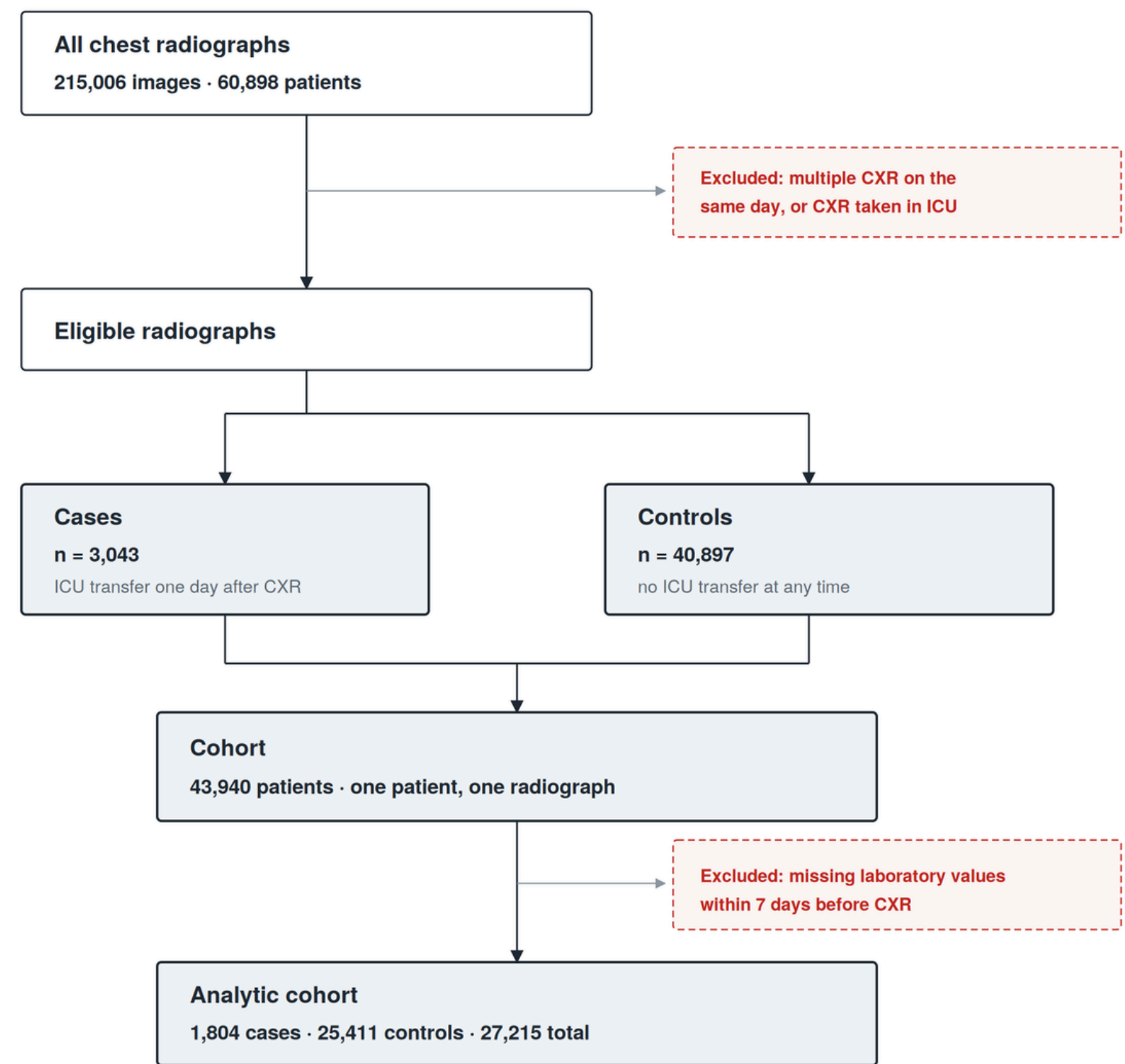
Clinical data + CXR

Novelty: define patient phenotypes without CXR or outcome information, then ask where CXR adds the most predictive value.

This avoids turning the study into another overall clinical-vs-multimodal benchmark.

Cohort construction

One patient, one eligible radiograph; non-ICU index CXR



Current analytic cohort

27,215

patients / radiographs

Cases 1,804

Controls 25,411

Event rate 6.6%

Cohort figure and counts supplied from the current preprocessing output.

What is currently in the analytic dataset?

27,215 rows × 40 columns in the supplied preprocessed cohort

Demographics

age, sex

CBC

Hgb, Hct, WBC, platelet, RBC,
MCHC, MCH, MCV, RDW

Differential

neutrophil %, lymphocyte %

Chemistry / renal

Na, K, Cl, bicarbonate,
anion gap, BUN, creatinine

Proposed analysis workflow

Train the phenotypes and prediction models without leaking information from the test set



Train n=1,960 · Validation n=420 · Test n=420
Each dataset: Case 50% · Control 50%

All preprocessing parameters — imputation, scaling, feature selection, CXR projection and cluster centroids — are learned from Train only.

Validation selects hyperparameters / K; Test is used once for final overall and phenotype-stratified evaluation.

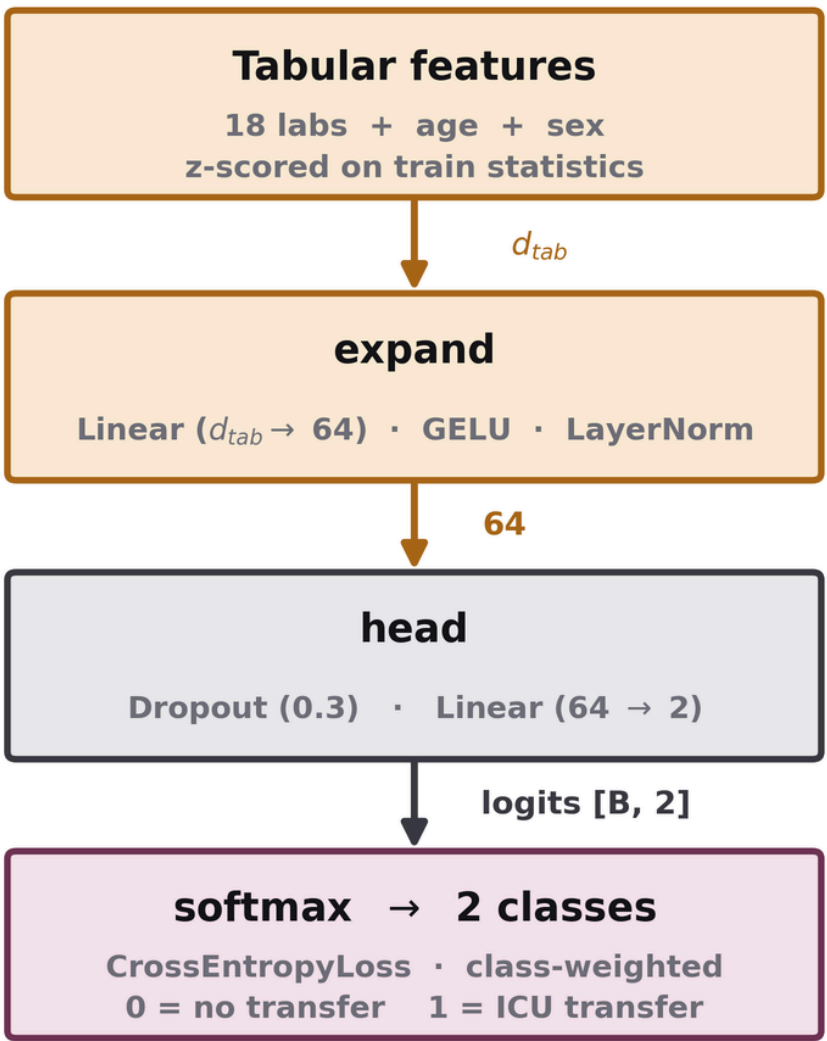
Model architecture

Controlled comparison: Model B adds a frozen RAD-DINO CXR branch

Model A uses tabular clinical data only; Model B keeps the same tabular branch and adds CXR features through late fusion.

Model A · Tabular-only baseline

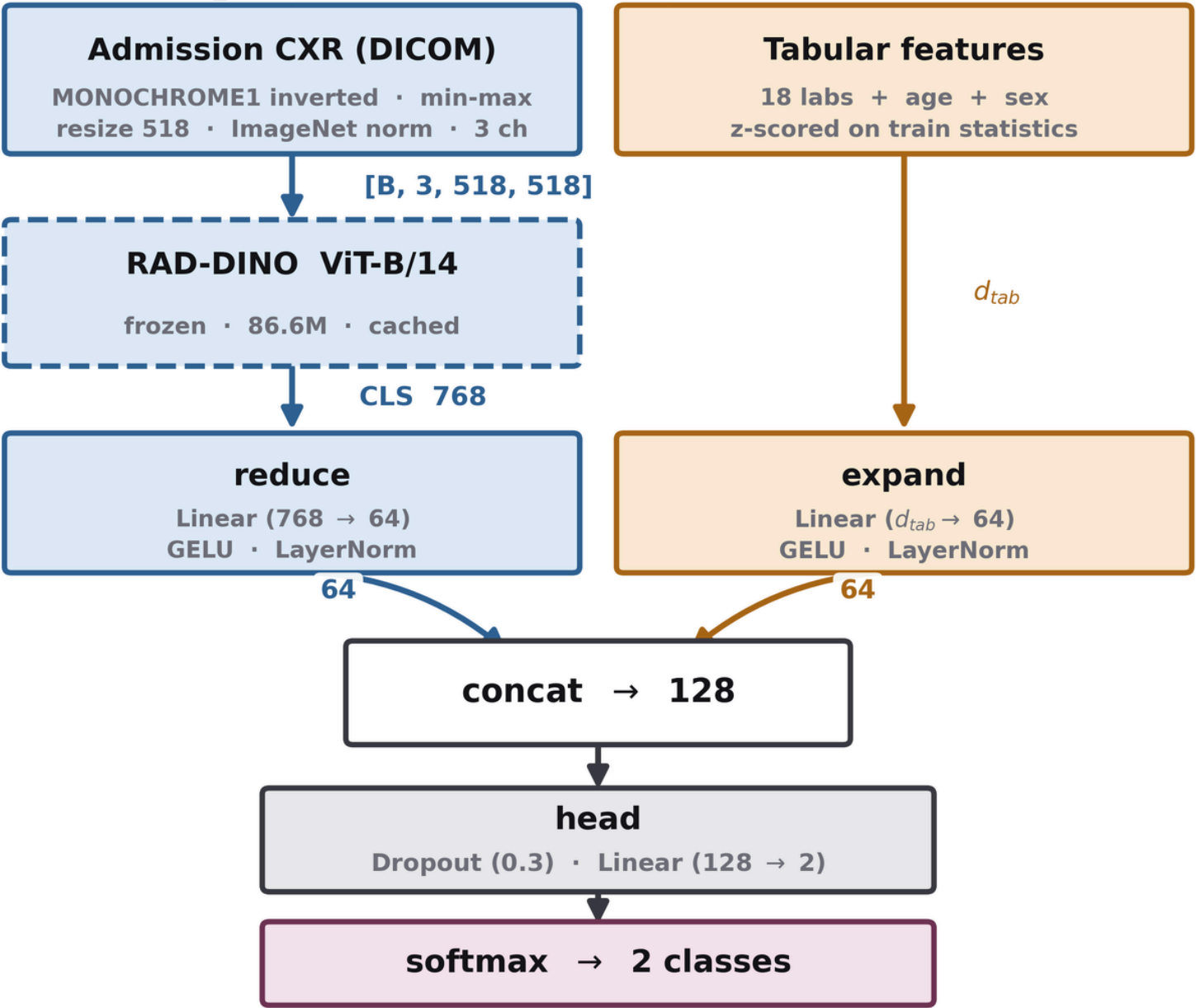
use_image = False · laboratory + demographics



How much signal the structured data alone carries.

Model B · RAD-DINO fusion

use_image = True · frozen image encoder + late fusion



Key point: Model B estimates the incremental value of adding CXR representation to the same tabular baseline.

Modeling and evaluation

Keep Model A and B identical except for the CXR feature

Model A Tabular-only baseline
(18 laboratory variables + age + sex)

Tabular MLP

Model B Clinical + CXR
Model A inputs + index CXR

Tabular MLP + frozen RAD-DINO

Primary metric

AUPRC

Secondary

AUROC · sensitivity · specificity
PPV/NPV · Brier · calibration

Comparison

Δ AUPRC and Δ AUROC
paired bootstrap 95% CI

Key inference: test whether the magnitude of Model B – Model A differs across phenotypes — not merely whether one cluster is ‘significant’ and another is not.

How would we interpret the results?

The clinically interesting result is heterogeneity, not necessarily a large overall ΔAUC

A	Large gain in one phenotype	CXR contains complementary prognostic information concentrated in a clinically defined subgroup.	Most informative
B	Similar gain in all phenotypes	CXR contributes broadly across the cohort.	Supports general multimodal use
C	Little or no gain	Clinical data already contain most predictive signal, or imaging representation is insufficient.	Negative but informative
D	Unstable cluster estimates	Too few ICU-transfer events or weak cluster reproducibility.	Exploratory only

Caveat: because the cohort contains patients who already had a CXR, this study cannot conclude that a specific phenotype ‘does not need a CXR’. It evaluates predictive incremental value.

Immediate next steps

What should be fixed before model training starts?

1 Freeze outcome window

Current extraction uses next-day ICU transfer; decide 24 h vs 48 h a priori.

2 Finalize clinical variables

Confirm availability/timing of vitals, SpO₂ and oxygen support; document missingness.

3 Define split

Patient-level Train / Validation / Test before any scaling, PCA or clustering.

4 Check cluster feasibility

Estimate ICU-transfer events per candidate K; likely start with K=2–3.

5 Extract CXR features

Use a frozen encoder; fit any dimensionality reduction on Train only.

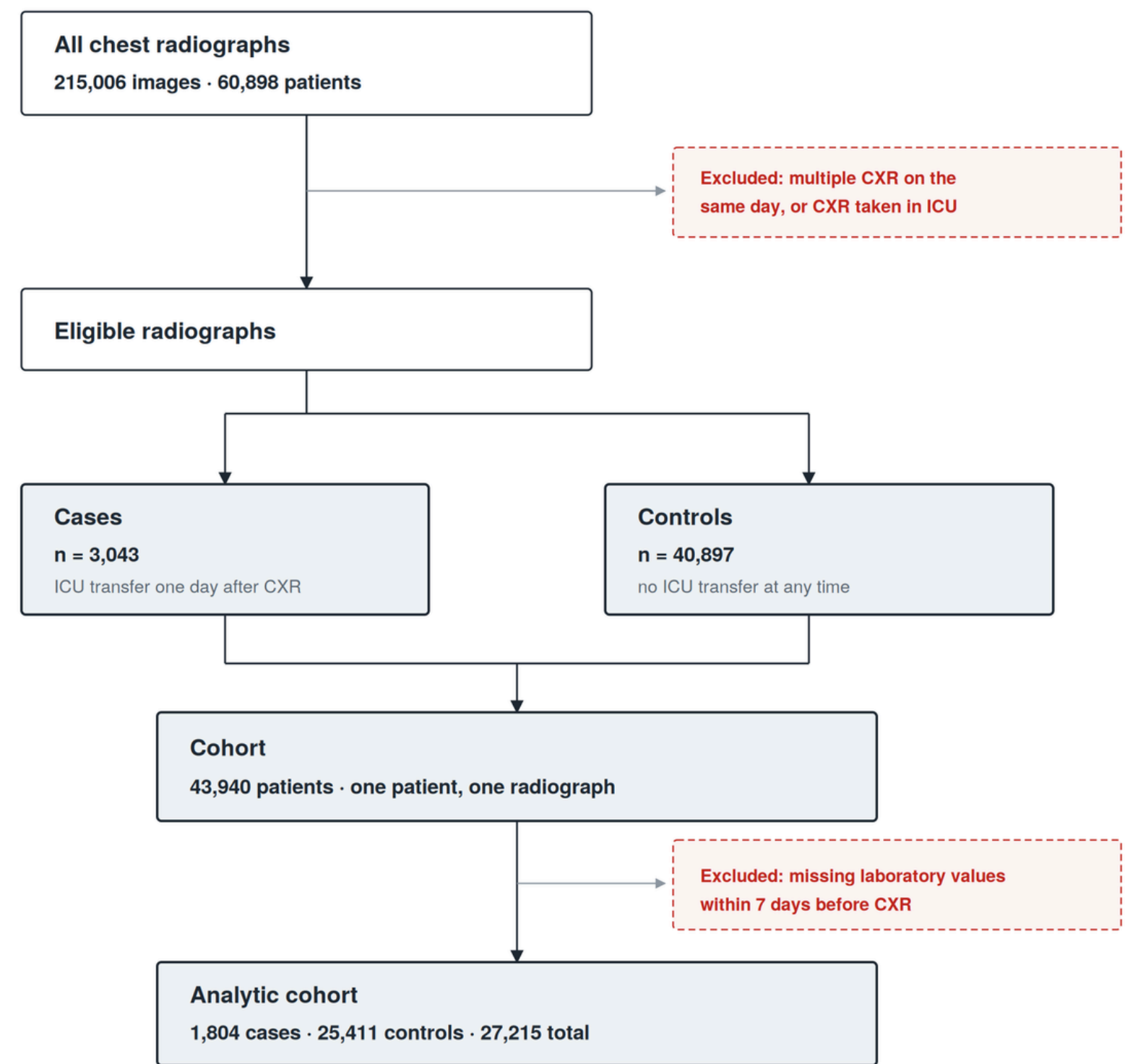
6 Pre-specify statistics

Primary AUPRC; paired bootstrap CI for overall and phenotype-specific Δ performance.

Goal: a reproducible comparison in which the only difference between Model A and Model B is the information contributed by CXR.

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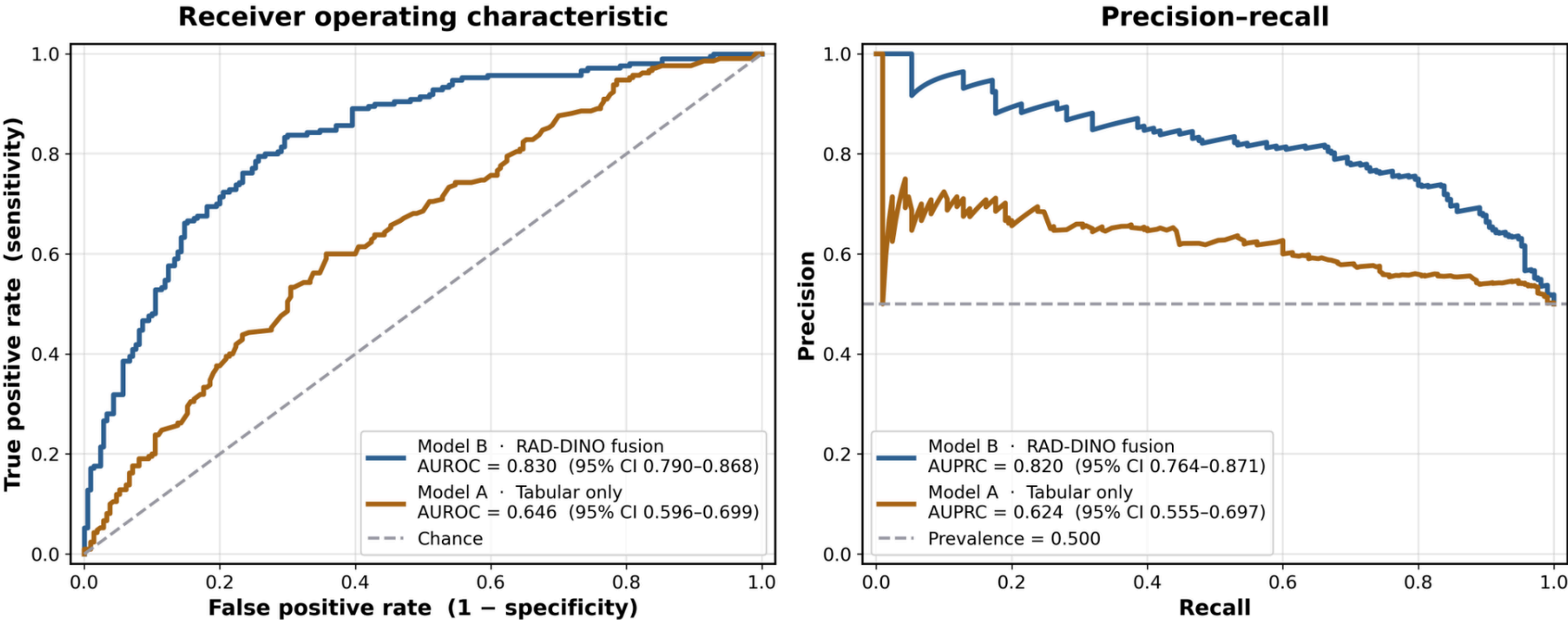
Event rate 6.6%

Cohort figure and counts supplied from the current preprocessing output.

Adding CXR substantially improves ICU transfer prediction

Initial held-out analysis before phenotype stratification: Model A (tabular only) vs Model B (tabular + CXR)

ICU transfer prediction · held-out test set (n = 420, 210 pos / 210 neg)



Overall Δ AUROC +0.183 and Δ AUPRC +0.193 → next question: does this benefit differ across clinical phenotypes?

Why stratify patients into clinical phenotypes?

K-means is used to test whether the incremental value of CXR is heterogeneous across patient subgroups

Study logic

Overall cohort



K-means using clinical variables

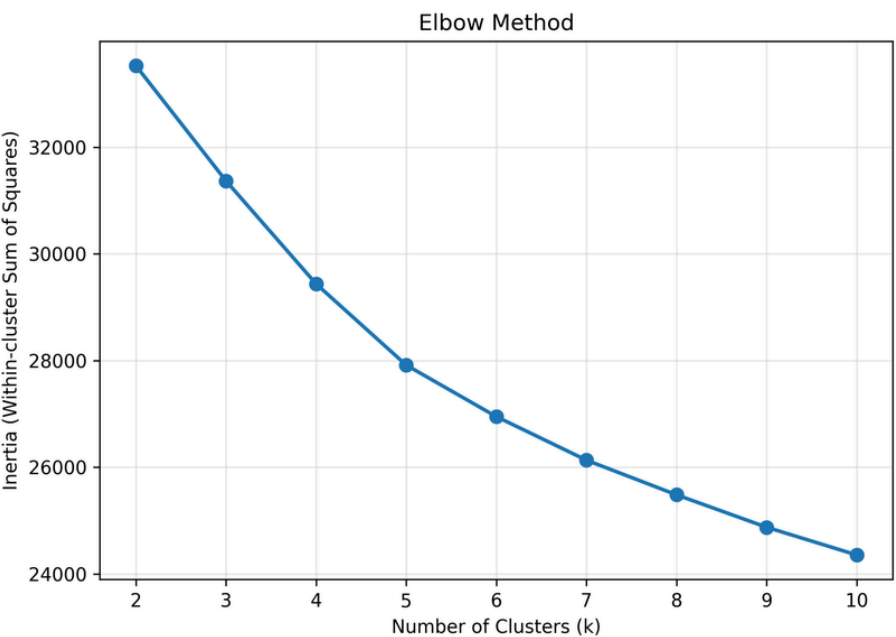
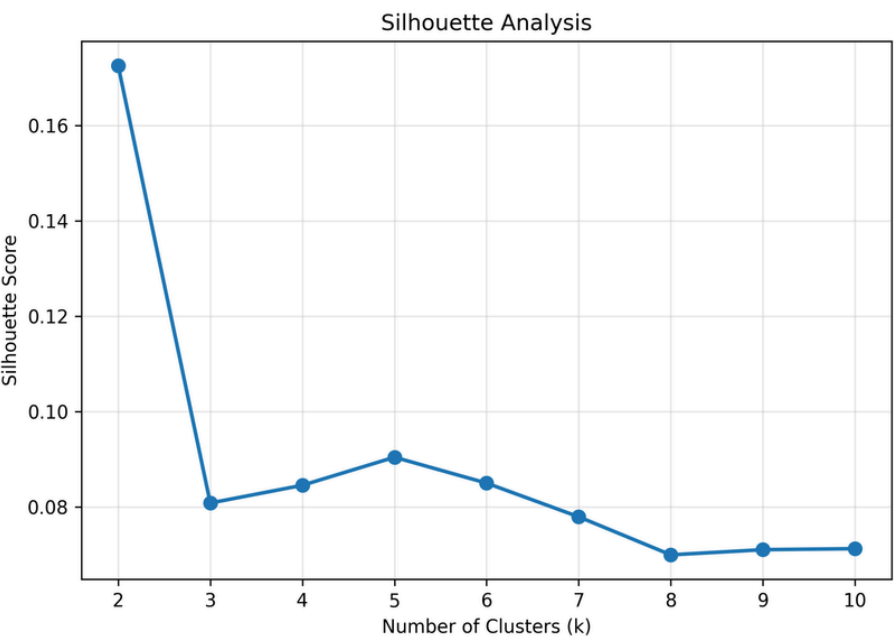


4 clinical phenotypes



Compare Model A vs B
within each phenotype

Goal: identify who benefits most from
adding CXR—not merely whether CXR
improves prediction overall.



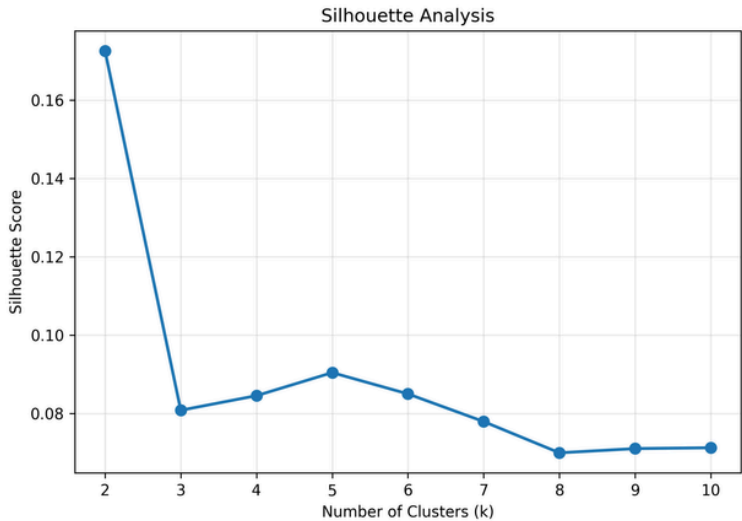
Why k = 4 despite k = 2 having the strongest separation?

k = 2 likely reflects a broad severity axis (e.g., relatively healthier vs sicker patients). That split may be statistically cleaner but is too coarse for our clinical question. We therefore chose k = 4 as a pragmatic balance between subgroup granularity, sample size, and clinical interpretability, allowing us to examine heterogeneous CXR benefit across distinct phenotypes.

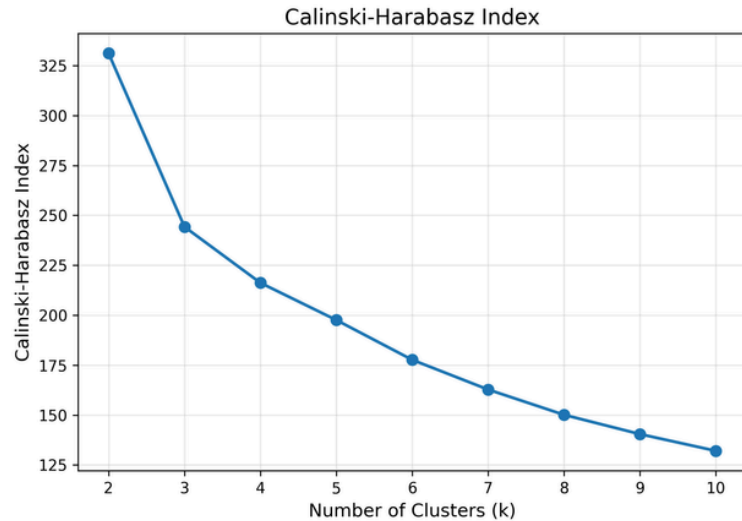
Selecting k = 4: clustering diagnostics

No single internal metric defines the clinically optimal k; diagnostics were interpreted together with phenotype utility

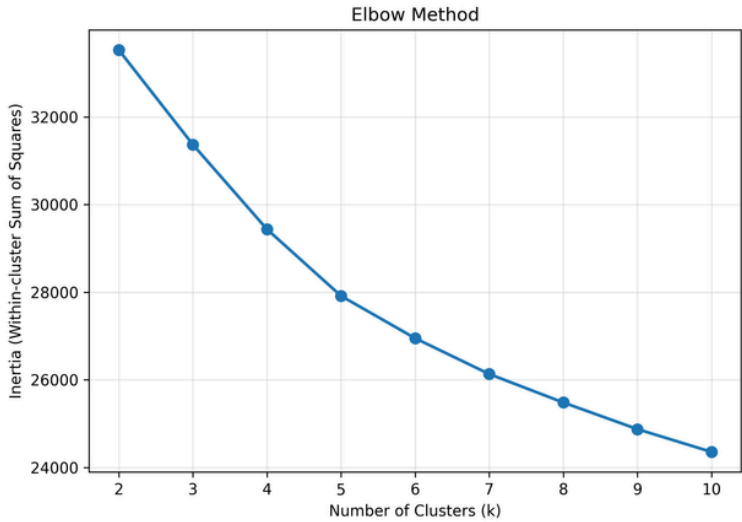
Silhouette



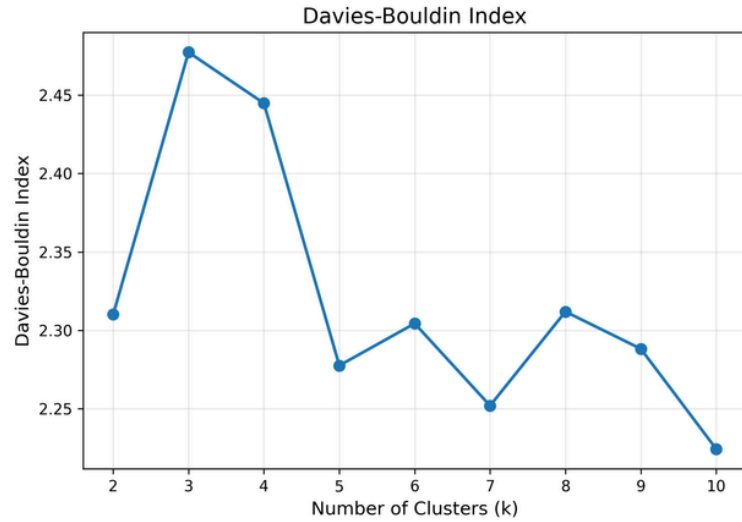
Calinski-Harabasz



Elbow / inertia



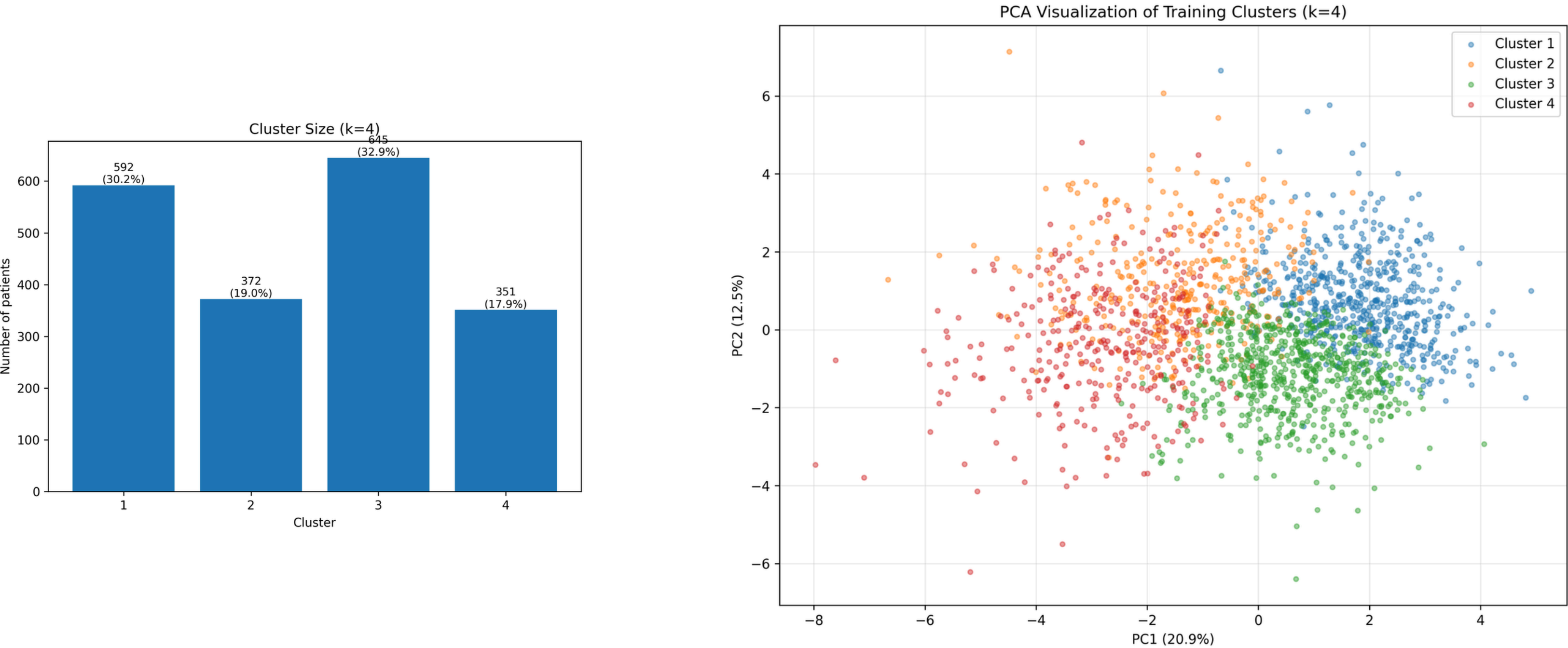
Davies-Bouldin



Interpretation: k=2 is geometrically strongest; k=4 is intentionally selected for clinically useful phenotypic resolution.

k = 4 produces sizeable, partially separated patient groups

All four phenotypes retain sufficient representation for downstream subgroup-specific prediction analyses

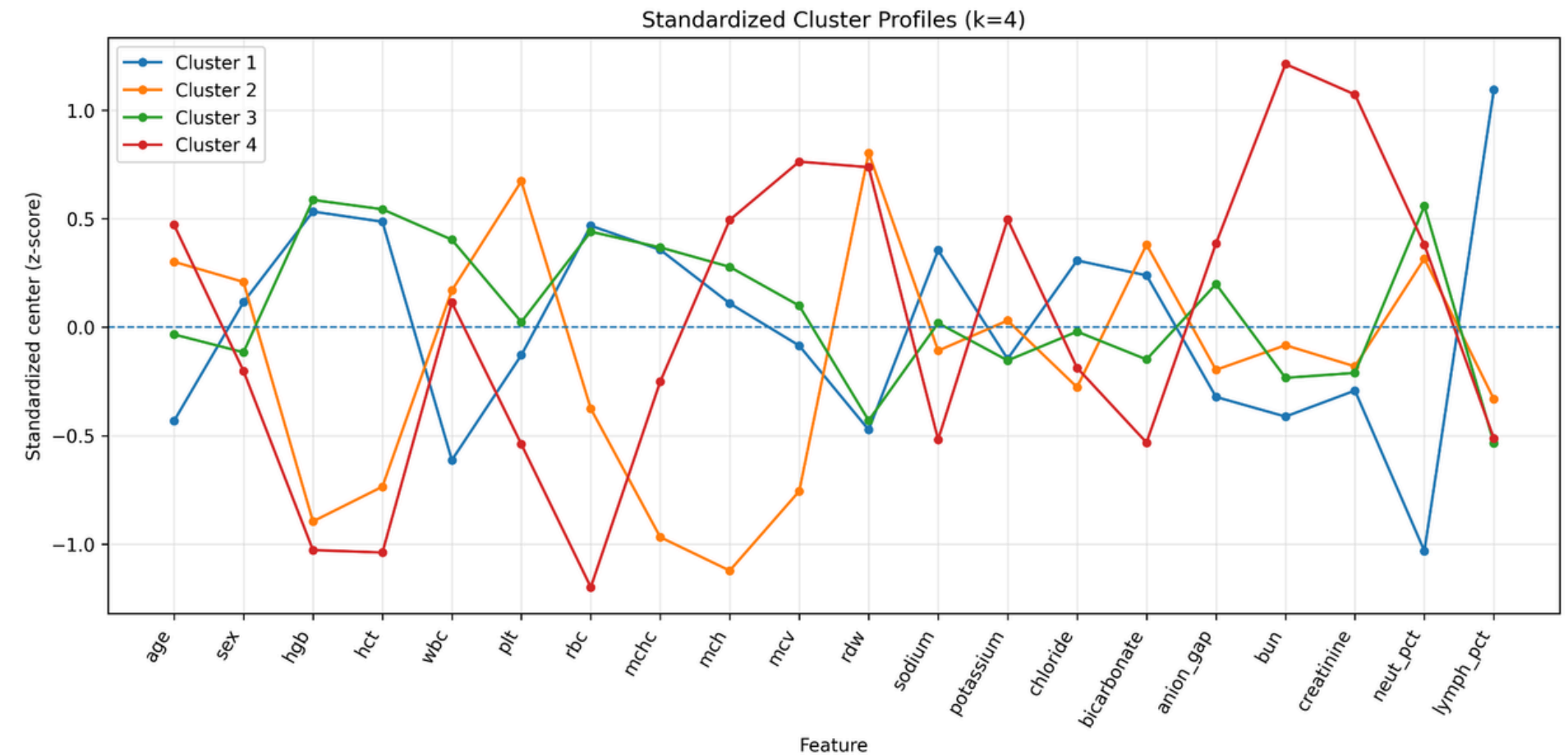
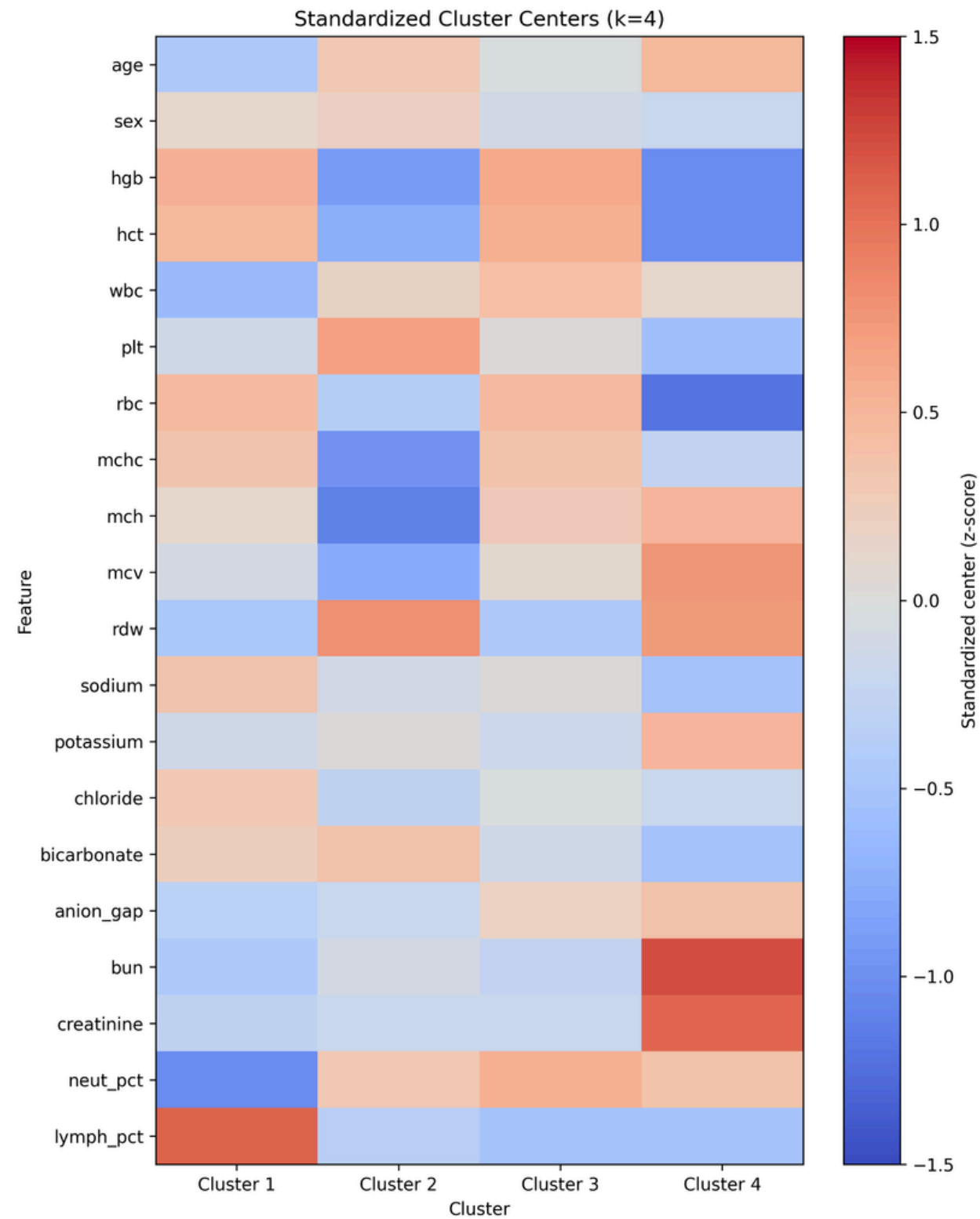


Cluster 1: 592 (30.2%) · Cluster 2: 372 (19.0%) · Cluster 3: 645 (32.9%) · Cluster 4: 351 (17.9%)

PCA is shown for visualization only; overlap is expected because K-means clustering was performed in the full standardized feature space.

Clinical characteristics of the four phenotypes

Standardized cluster centers reveal distinct hematologic, renal, inflammatory, and demographic patterns

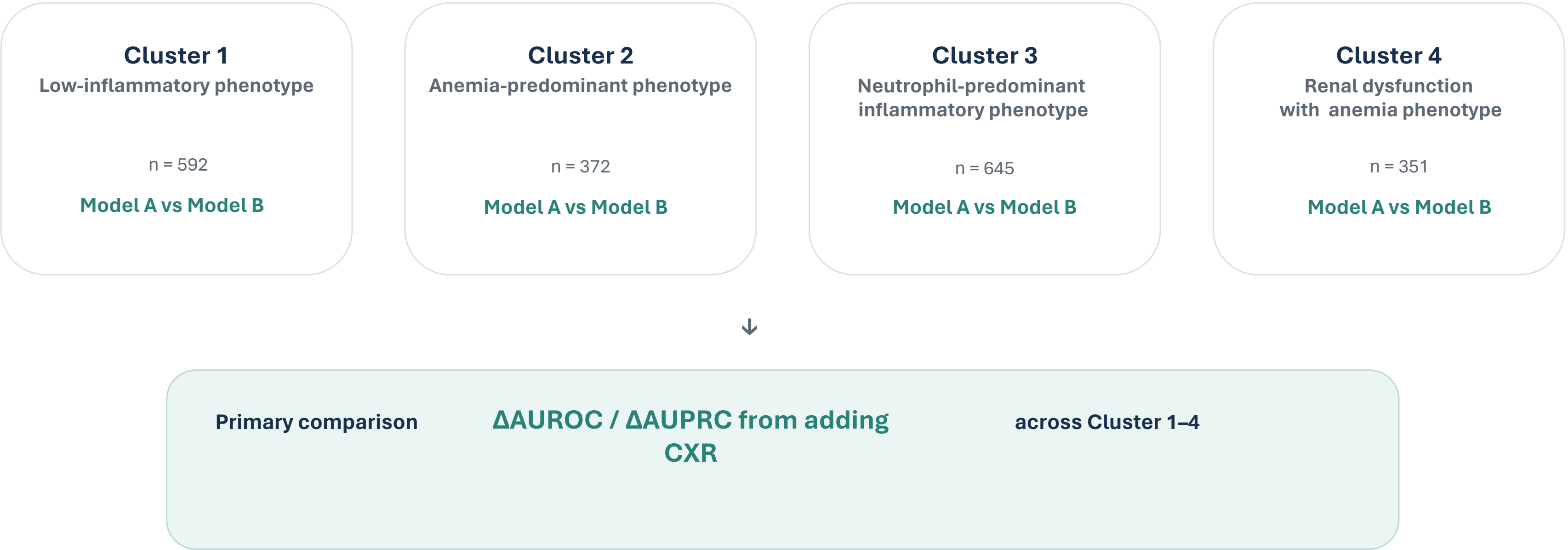


Descriptive phenotype interpretation

- Cluster 1: younger, preserved Hb/Hct, lymphocyte-predominant profile
- Cluster 2: anemia / thrombocytosis pattern with higher RDW
- Cluster 3: relatively preserved blood counts with neutrophil-predominant profile
- Cluster 4: older, anemia with markedly higher BUN/creatinine → renal dysfunction phenotype

Next analysis: phenotype-specific incremental value of CXR

The central endpoint is Δ performance between Model B and Model A within each of the four clinical phenotypes



Hypothesis: CXR will provide greater incremental information in selected phenotypes rather than uniformly across all patients.

CXR provides the greatest incremental value in the inflammatory phenotype

Phenotype-specific improvement of Model B (clinical + CXR) over Model A (clinical only)

Cluster	n	Δ AUROC	Δ AUPRC
1 Low-inflammatory phenotype	129	+0.135	+0.179
2 Anemia-predominant phenotype	81	+0.262	+0.172
3 Neutrophil-predominant inflammatory phenotype	125	+0.251	+0.239
4 Renal dysfunction with anemia phenotype	85	+0.255	+0.106
ALL	420	+0.184	+0.196

Key finding

Cluster 3
Inflammatory phenotype

Δ AUPRC = +0.239

Largest improvement among
all four phenotypes

Interpretation

Adding CXR appears particularly informative in the inflammatory phenotype, supporting heterogeneous incremental value of imaging across clinical phenotypes.

Note: between-cluster differences should be confirmed with confidence intervals / bootstrap comparison before claiming statistical heterogeneity.

References

Key literature supporting variable selection, multimodal modeling and phenotyping

1. Churpek MM, et al. Using electronic health record data to develop and validate a prediction model for adverse outcomes in the wards. Crit Care Med. 2014;42:841–848. doi:10.1097/CCM.0000000000000038.
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8. Hurley NC, et al. Visualization of emergency department clinical data for interpretable patient phenotyping. Smart Health. 2022;25:100285. doi:10.1016/j.smhl.2022.100285.
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